

Index

1. Definitions
2. Methodology
3. Progress
4. Future steps

Definitions

Primary unit: A sentence, a bullet point, or a row in a table. For figures, only the title of the figure is used. These were extracted manually since extracting them is an objective endeavour suitable for human annotators.

Recommendation: A clause extracted from the specifications according to the methodology devised by the human annotator given the role of extractor.

Cohen's kappa agreement: Calculated on whether or not a primary unit contains a recommendation or not. If it does, a 1 is assigned to the primary unit. If it does not, a 0 is assigned.

Methodology

Stage 1. Two annotators (extractor and validator) independently extracted recommendations from three specifications. Pilot studies were conducted to refine the method and resolve discrepancies. This process was repeated until a strong agreement was achieved (Cohen's Kappa ≥ 0.75).

Stage 2. The same sections' recommendations were then extracted by NotebookLM using the established methodology. The agreement between the manual extractor and NotebookLM exceeded 0.75, confirming its reliability. The approach was further validated on 30% of the specifications, supporting the suitability of automated extraction.

Stage 3. NotebookLM was used to extract recommendations from the complete specifications.

Progress

Pilot stage 1

Pilot stage 2

Validation

Completion

spdx

0.46

0.82

0.75

done

cyclonedx

0.454

0.895

0.650

cisa

0.58

0.779

0.803

done

Future steps

- For the dataset: prior literature already provides datasets with publicly available sboms [1]. Additionally, we can also generate SBOMs ourselves using popular SBOM generation tools and test on those. We intend on a mix of both methods.
- For choosing what to test on: Tagging of the extracted recommendations with labels such as objective - local, objective - global, subjective, alongside the main keywords: MUST, SHOULD, CAN. Details on testing can be determined after this qualitative coding of the recommendations.
- For creating the test cases: A command line interface already present in tools such as sbomqs seems to be the best option for testing. The details on the nature of the data represented as a result of the test suite being run is to be determined after the qualitative coding of the recommendations.